

Group-Index Extraction with Unbalanced Mach–Zehnder Interferometers and Bragg-Grating Fabry–Pérot Cavities in 220-nm Silicon-on-Insulator

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Abstract—This proposal describes six quasi-TE test circuits in 220-nm silicon-on-insulator for extracting the group index of a 500-nm strip waveguide with two interferometric methods. Three Y-branch Mach–Zehnder interferometers with path-length differences of 50, 100 and 200 micrometres provide the group index across the grating-coupler band and show how the free spectral range scales with the imbalance. Three Fabry–Pérot cavities, each formed by two 200-period sidewall-corrugated Bragg mirrors separated by 100 micrometres, use grating periods of 307, 312 and 317 nm. Three-dimensional FDTD simulations of a single grating period give Bragg wavelengths of 1516, 1530 and 1545 nm and a coupling coefficient of about 54000 per meter. Combining the free spectral range of each cavity with its stopband width gives the group index at the Bragg wavelength. Combining it instead with the interferometer group index gives the mirror penetration depth and the coupling coefficient. Transfer-matrix simulations predict loaded quality factors of 220000 to 270000 and linewidths of 6 to 7 pm. Applied to spectra sampled at the 1-pm laser step, the extraction recovers the group index to within 0.002% from the interferometers and 0.4% from the cavities, and the coupling coefficient to within 2%.

Index Terms—Silicon photonics, Bragg grating, Fabry–Pérot cavity, Mach–Zehnder interferometer, group index.

I. INTRODUCTION

THE group index n_g sets the free spectral range (FSR) of every interferometer and resonator built from a waveguide, and it is one of the most sensitive measures of the fabricated waveguide cross-section. The standard way to measure it is an unbalanced Mach–Zehnder interferometer (MZI), whose fringe spacing gives n_g directly [1], [2]. A Fabry–Pérot (FP) cavity made from two waveguide Bragg gratings offers a complementary measurement. Its resonances are much narrower than MZI fringes, it measures n_g only inside the grating stopband, and its spectrum also carries information about the mirrors (coupling coefficient, reflectivity, penetration depth) and about the round-trip loss.

The objective of this design, as required by the course [1], is to extract n_g of the standard 500 nm $\times$ 220 nm strip waveguide (quasi-TE) from measurements and to compare it with mode-solver predictions. Two device families are used:

- 1) Three unbalanced MZIs with Y-branch splitters and $\Delta L = 50, 100$ and $200 \mu\text{m}$. They give $n_g(\lambda)$ over the ~ 50 nm grating-coupler window and test the expected scaling $\text{FSR} \propto 1/\Delta L$.
- 2) Three Bragg-mirror FP cavities with grating periods $\Lambda = 307, 312$ and 317 nm, all with a mirror gap of $L = 100 \mu\text{m}$. Each gives n_g at its Bragg wavelength

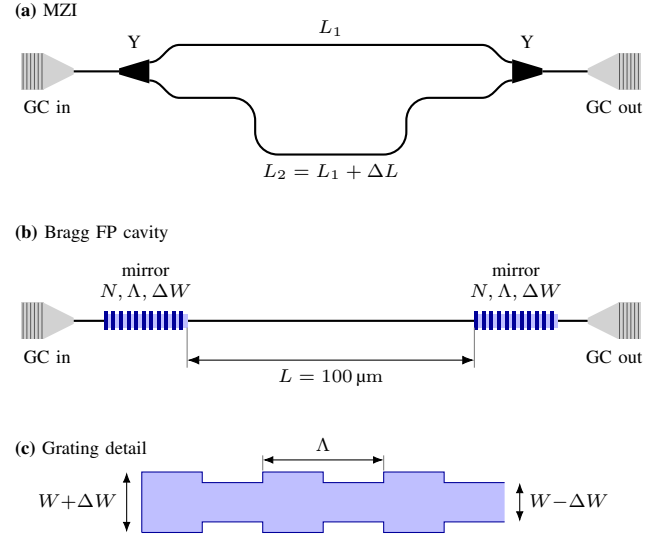


Fig. 1. Circuit schematics (not to scale). (a) Y-branch MZI with imbalance ΔL . (b) Fabry–Pérot cavity formed by two identical Bragg mirrors; $L = 100 \mu\text{m}$ is the gap between the inner grating edges. (c) Rectangular sidewall corrugation in the PDK convention (width $W \pm \Delta W$, each sidewall displaced by $\pm \Delta W/2$).

λ_B as a cross-check of the MZIs. Combined with the MZI value, each also gives the penetration depth and coupling coefficient of its mirrors. The period sweep moves λ_B across the band and gives the effective index of the grating.

The FP cavities also illustrate how the mirror strength must be chosen relative to the intracavity loss and the measurement resolution (Section III-D). Fig. 1 shows the circuits.

II. THEORY

A. Waveguide Compact Model

The waveguide is described by a second-order Taylor expansion of the effective index about $\lambda_0 = 1.55 \mu\text{m}$,

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad (1)$$

with propagation constant $\beta = 2\pi n_{\text{eff}}/\lambda$. The group index follows from the dispersion of n_{eff} ,

$$n_g(\lambda) = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} = n_1 - n_2\lambda_0 + n_3(\lambda_0^2 - \lambda^2), \quad (2)$$

so that $n_g(\lambda_0) = n_1 - n_2\lambda_0$.

B. Mach-Zehnder Interferometer

An ideal Y-branch [3] splits the input field into two arms with amplitude $1/\sqrt{2}$ each, and the output Y-branch recombines them with another factor $1/\sqrt{2}$. With arm lengths L_1 and $L_2 = L_1 + \Delta L$ and a power attenuation coefficient α , the output field is

$$\frac{E_o}{E_i} = \frac{1}{2} \left[e^{-i\beta L_1 - \alpha L_1/2} + e^{-i\beta L_2 - \alpha L_2/2} \right], \quad (3)$$

and the transmitted power is

$$T_{\text{MZI}}(\lambda) = \frac{e^{-\alpha L_1}}{4} \left[1 + e^{-\alpha \Delta L} + 2e^{-\alpha \Delta L/2} \cos(\beta \Delta L) \right]. \quad (4)$$

For negligible loss, $T_{\text{MZI}} = \frac{1}{2} [1 + \cos(\beta \Delta L)]$. The measured spectrum is additionally multiplied by the two grating-coupler responses, which vary slowly with λ and are removed by normalization.

C. Group Index from the MZI Free Spectral Range

Transmission maxima occur at $\beta(\lambda_m)\Delta L = 2\pi m$. For two adjacent maxima,

$$[\beta(\lambda_m) - \beta(\lambda_{m+1})] \Delta L = 2\pi. \quad (5)$$

Expanding β to first order about the midpoint λ of the two peaks,

$$\frac{d\beta}{d\lambda} = -\frac{2\pi}{\lambda^2} \left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right) = -\frac{2\pi n_g}{\lambda^2}, \quad (6)$$

gives $|d\beta/d\lambda| \text{FSR} \Delta L = 2\pi$, i.e.

$$\text{FSR}(\lambda) = \frac{\lambda^2}{n_g \Delta L} \implies n_g(\lambda) = \frac{\lambda^2}{\Delta L \cdot \text{FSR}(\lambda)}. \quad (7)$$

Each pair of adjacent peaks gives one value of n_g , so a single spectrum yields $n_g(\lambda)$ across the measurement band.

D. Sidewall-Corrugated Bragg Mirror

The mirrors are first-order rectangular gratings in which the waveguide width alternates between $W + \Delta W$ and $W - \Delta W$ every half period. This is the convention of the EBeam PDK cell `ebeam_bragg_te1550` [4], in which each sidewall is displaced by $\pm \Delta W/2$. The Bragg condition is

$$\lambda_B = 2\bar{n}\Lambda, \quad (8)$$

where $\bar{n}$ is the average effective index of the two segments. In coupled-mode theory [5], [6], a uniform grating of length $L_g = N\Lambda$ and coupling coefficient κ has the amplitude reflectivity

$$r(\delta) = \frac{-i\kappa \sinh(sL_g)}{s \cosh(sL_g) + i\delta \sinh(sL_g)}, \quad s^2 = \kappa^2 - \delta^2, \quad (9)$$

with detuning $\delta = \beta - \pi/\Lambda$. The peak power reflectivity, stopband width (between the band edges $\delta = \pm\kappa$) and penetration depth are [2]

$$R_{\text{max}} = \tanh^2(\kappa L_g), \quad (10)$$

$$\Delta\lambda_{\text{SB}} = \frac{\lambda_B^2 \kappa}{\pi n_g}, \quad (11)$$

$$L_p = \frac{\tanh(\kappa L_g)}{2\kappa}. \quad (12)$$

L_p is the one-way distance that, travelled at n_g , reproduces the group delay of the reflection at λ_B . For a rectangular corrugation, a first estimate of κ is obtained from the effective indices of the wide and narrow waveguides, $\kappa \approx 2\Delta n/\lambda_B$ with $\Delta n = n_{\text{eff}}(W + \Delta W) - n_{\text{eff}}(W - \Delta W)$ [2].

E. Bragg Fabry-Pérot Cavity

Two identical mirrors with power reflectivity $R = |r|^2$ and transmissivity $1 - R$ are separated by a strip waveguide of length L (inner grating edge to inner grating edge). With single-pass power transmission $a = e^{-\alpha L}$, summing the multiply reflected fields gives the Airy transmission

$$T_{\text{FP}}(\lambda) = \frac{(1 - R)^2 a}{(1 - Ra)^2 + 4Ra \sin^2(\theta/2)}, \quad (13)$$

with round-trip phase $\theta = 2\beta L + 2\phi_r$, where ϕ_r is the mirror reflection phase. The resonances have finesse F , linewidth $\delta\lambda$, loaded quality factor Q_L and peak transmission T_{pk} :

$$F = \frac{\pi\sqrt{Ra}}{1 - Ra}, \quad \delta\lambda = \frac{\text{FSR}}{F}, \quad Q_L = \frac{\lambda}{\delta\lambda}, \quad (14)$$

$$T_{\text{pk}} = \frac{(1 - R)^2 a}{(1 - Ra)^2}. \quad (15)$$

Equation (15) sets the design trade-off. For mirror transmission $1 - R \ll 1 - a$ (an under-coupled cavity), T_{pk} falls as $(1 - R)^2$ and the resonances disappear into the noise floor. For $1 - R \gg 1 - a$, the resonances are bright but broad.

F. Group Index and Loss from the FP Spectrum

Adjacent resonances satisfy $\theta(\lambda_m) - \theta(\lambda_{m+1}) = 2\pi$. Besides the waveguide dispersion $d\beta/d\lambda$, the round-trip phase contains the reflection-phase dispersion of the two mirrors. Near λ_B , each mirror reflection adds a round-trip group delay $2n_g L_p/c$ [Eq. (12)], so the round-trip group delay is $\tau_{\text{RT}} = 2n_g(L + 2L_p)/c$. Following the steps that led to Eq. (7),

$$\text{FSR} = \frac{\lambda_B^2}{2n_g(L + L_0)}, \quad L_0 = 2L_p. \quad (16)$$

If L_p is known, a single cavity gives n_g . For the strong mirrors used here, $\tanh(\kappa L_g) \approx 1$, so $L_p \approx 1/2\kappa$. With κ from the stopband width, Eq. (11), this is $L_p \approx \lambda_B^2/(2\pi n_g \Delta\lambda_{\text{SB}})$. Substituting into Eq. (16) and solving for n_g gives

$$n_g(\lambda_B) = \frac{\lambda_B^2}{L} \left(\frac{1}{2\text{FSR}} - \frac{1}{\pi \Delta\lambda_{\text{SB}}} \right), \quad (17)$$

which uses only the measured FSR at λ_B and the measured stopband width. Conversely, with n_g taken from the MZIs, Eq. (16) gives the penetration depth directly,

$$L_p = \frac{1}{2} \left(\frac{\lambda_B^2}{2n_g \text{FSR}} - L \right), \quad (18)$$

and hence κ through Eq. (12), independently of the stopband width.

Because L_p grows toward the band edges, the resonance spacing is largest at λ_B and decreases on both sides. The FSR at λ_B is therefore obtained by fitting the spacings of adjacent resonance pairs with an even function $a + b(\lambda - \lambda_B)^2$ and

taking a . Because $2L_p$ is about 16% of the effective cavity length, a 1% error in $\Delta\lambda_{\text{SB}}$ changes the n_g from Eq. (17) by about 0.16%. The cavities therefore cross-check n_g , while the MZIs measure it precisely.

For the loss, Eq. (14) is inverted for the round-trip amplitude factor $\rho = Ra$ of each cavity,

$$\rho = \frac{\pi^2 + 2F^2 - \pi\sqrt{\pi^2 + 4F^2}}{2F^2}, \quad (19)$$

and

$$-\ln \rho = -\ln R + \alpha L \quad (20)$$

With $R = \tanh^2(\kappa L_g)$ from Eq. (10), this gives the propagation loss α . The estimate is coarse, because $1 - R \approx 4e^{-2\kappa L_g}$ is very sensitive to κ .

Finally, the period sweep gives the grating effective index directly from Eq. (8), $\bar{n} = \lambda_B/2\Lambda$. Differentiating Eq. (8) gives

$$\frac{d\lambda_B}{d\Lambda} = \frac{2\bar{n}^2}{n_g}, \quad (21)$$

which links the effective and group indices and serves as a consistency check between the measured λ_B and n_g .

III. MODELLING AND SIMULATION

All simulations use Tidy3D. Waveguide modes come from its finite-difference eigenmode (FDE) solver; the Bragg grating comes from three-dimensional FDTD simulations of a single grating period. The circuit responses are computed from the analytic transfer functions of Section II and a transfer-matrix (TMM) model of the cavities, evaluated in Python with the simulated waveguide and grating parameters.

A. Waveguide Geometry and Mode Profile

All devices use the quasi-TE mode of a 500 nm-wide, 220 nm-thick silicon strip on a buried oxide with oxide cladding. Table I lists the common design parameters. The width is single-mode at 1550 nm and matches the PDK grating couplers, Y-branch and Bragg cell, so no tapers are needed. Fig. 2 shows the mode profile, computed with a 5 nm mesh over the core. In a convergence test, n_{eff} changed by 0.015 between 10 and 5 nm mesh and by less than 3×10^{-4} between 5 and 2.5 nm. Materials are crystalline Si and SiO₂ (Palik, $n = 1.444$ at 1550 nm).

B. Effective and Group Index, Compact Model

The mode solver gives $n_{\text{eff}} = 2.444$, $n_g = 4.192$ and a TE polarization fraction of 0.98 at 1550 nm. A quadratic fit of $n_{\text{eff}}(\lambda)$ over 1500–1600 nm (11 wavelengths, fit residual $< 2 \times 10^{-6}$) gives the compact model

$$n_{\text{eff}}(\lambda) = 2.4444 - 1.1271(\lambda - \lambda_0) - 0.0489(\lambda - \lambda_0)^2, \quad (22)$$

with λ in μm . Through Eq. (2), this model reproduces the solver's directly computed group index to within 5×10^{-4} ($n_g = 4.184$, 4.191 and 4.199 at 1500, 1550 and 1600 nm; Fig. 3). The group index changes by only ≈ 0.015 across the band. Fabrication matters more: n_g at 1550 nm is 4.228 and

TABLE I
COMMON DESIGN PARAMETERS

Parameter	Value
Si thickness H / width W	220 nm / 500 nm
Polarization	quasi-TE (TE ₀)
Cladding	SiO ₂
Grating couplers	ebeam_gc_te1550, 127 μm pitch
Splitter (MZI)	ebeam_y_1550
MZI imbalance ΔL	50, 100, 200 μm
Bragg cell	ebeam_bragg_te1550, rectangular
Corrugation ΔW	40 nm (width 460/540 nm)
Periods per mirror N	200 ($L_g = 61.4\text{--}63.4 \mu\text{m}$)
Grating period Λ	307, 312, 317 nm
Mirror gap L	100 μm
<i>Simulated / assumed</i>	
n_{eff} , n_g at 1550 nm	2.444, 4.192 (FDE)
Coupling coefficient κ	$5.26\text{--}5.42 \times 10^4 \text{ m}^{-1}$ (FDTD)
Propagation loss α	1.4 dB/cm (assumed [4])

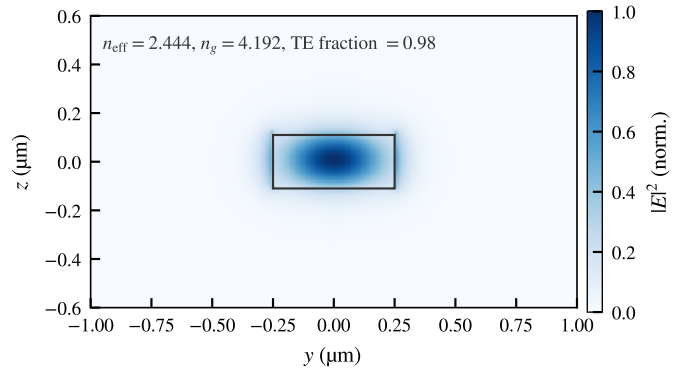


Fig. 2. Simulated quasi-TE mode profile ($|E|^2$) of the 500 nm $\times$ 220 nm strip waveguide at $\lambda = 1550$ nm (Tidy3D FDE). The rectangle marks the Si core.

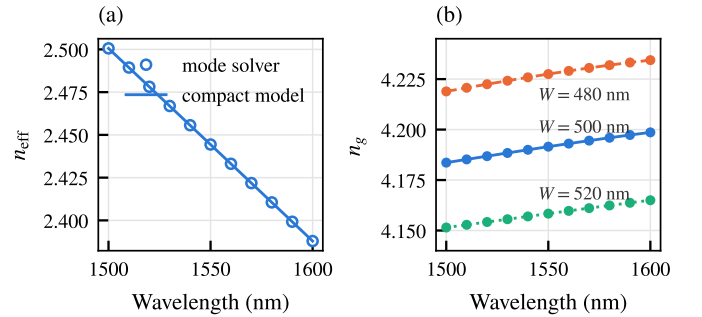


Fig. 3. (a) Effective index of the TE₀ mode from the mode solver (symbols) and the compact model of Eq. (22) (line). (b) Group index for widths of 480, 500 and 520 nm.

4.158 for widths of 480 and 520 nm, about 1.7×10^{-3} per nm of width. The measured n_g is therefore a sensitive check of the fabricated width.

C. MZI Design

The three imbalances, $\Delta L = 50$, 100 and 200 μm , give FSRs of 11.46, 5.73 and 2.87 nm at 1550 nm [Eq. (7)], i.e., about 4, 9 and 17 fringes in a 50 nm window. All three FSRs

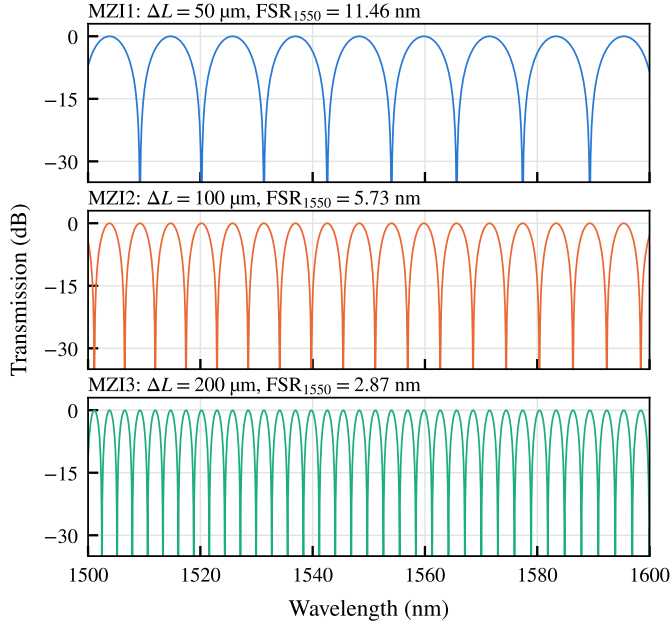


Fig. 4. Simulated transmission of the three MZIs [Eq. (4) with the compact model of Eq. (22)], excluding the grating-coupler response.

are far below the measurement bandwidth and far above the 1 pm laser step. The shortest imbalance gives few, widely spaced fringes that are least affected by fringe-position noise. The longest gives the densest sampling of $n_g(\lambda)$. Together they test the $1/\Delta L$ scaling of Eq. (7). Fig. 4 shows the spectra from Eq. (4) with the compact model and 1.4 dB/cm loss.

D. Bragg Mirror Design

With the PDK convention, $\Delta W = 40$ nm means that the waveguide width alternates between 540 and 460 nm. The coupling coefficient and the Bragg wavelength were obtained from 3D FDTD simulations of a single grating period, using Bloch boundary conditions at $k = \pi/\Lambda$. The two TE_0 band-edge modes, $f_1 < f_2$, were excited by broadband dipoles, and their frequencies were extracted by harmonic inversion of the ringing fields. Then $\lambda_B = 2c/(f_1 + f_2)$, $\Delta\lambda_{\text{SB}} = c/f_1 - c/f_2$ and $\kappa = \pi n_g(f_2 - f_1)/c$ [Eq. (11)]. The results for $\Lambda = 307$, 312 and 317 nm are:

- $\lambda_B = 1515.7, 1530.2$ and 1544.5 nm,
- $\Delta\lambda_{\text{SB}} = 9.18, 9.54$ and 9.83 nm,
- $\kappa = 5.26, 5.36$ and $5.42 \times 10^4 \text{ m}^{-1}$.

The stopbands shift by 2.88 nm per nm of period, in agreement with $2\bar{n}^2/n_g = 2.87$ from Eq. (21) with $\bar{n} = \lambda_B/2\Lambda = 2.452$. As expected, κ is nearly independent of the period. The simple effective-index estimate $\kappa \approx 2\Delta n/\lambda_B$ ($\Delta n = 0.128$ between the 540 and 460 nm strips) gives $1.65 \times 10^5 \text{ m}^{-1}$, three times larger. This confirms that the short segments ($\Lambda/2 \approx 155$ nm) are far too short for the mode to adapt to each width, so the full 3D simulation is needed.

The number of periods is chosen with Eq. (15) and the 1 pm laser step in mind. With $N = 200$ ($L_g \approx 62 \mu\text{m}$), $\kappa L_g \approx 3.3$ gives $R = 0.994$ – 0.996 and $L_p = 9.2$ – $9.5 \mu\text{m}$ [Eqs. (10), (12)]. The mirror transmission, $1 - R \approx 5 \times 10^{-3}$, is comparable to the single-pass loss (3×10^{-3} for $L = 100 \mu\text{m}$), so the

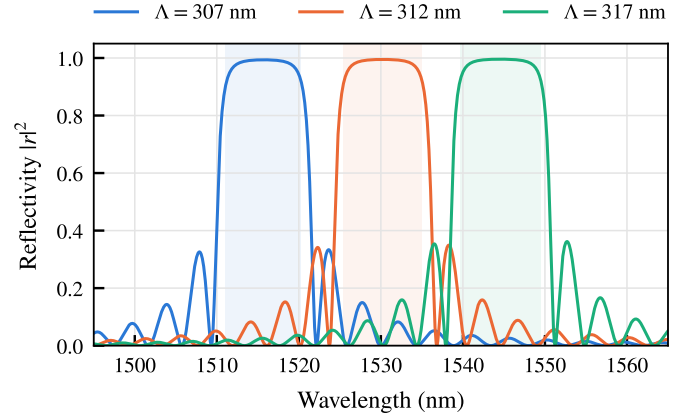


Fig. 5. Reflectivity of the three 200-period Bragg mirrors from the transfer-matrix model with λ_B and κ from the 3D FDTD unit-cell simulations. Shaded bands mark the FDTD stopbands.

cavities are close to critical coupling. This gives narrow but still visible resonances: linewidths of 5.7–7.0 pm, $Q_L \approx 2.2$ – 2.7×10^5 and $T_{\text{pk}} \approx -3.6$ to -4.9 dB (Table II). The linewidth spans six to seven laser steps, which is enough for a Lorentzian fit [Fig. 6(d)]. A higher fabricated propagation loss would dim and broaden the resonances. All three cavities use the same N , ΔW and L . Fig. 5 shows the mirror reflectivity.

E. Circuit Simulations

The FP cavities are modelled by a transfer-matrix method. Each mirror is a one-dimensional stack of $N = 200$ periods, with an index step $\Delta n = \kappa\lambda/2$ (so that the stack has the FDTD coupling coefficient) and an average index offset so that the stack reproduces the FDTD λ_B . The cavity is a strip section described by Eq. (22) with 1.4 dB/cm loss. Fig. 6 shows the three spectra and a single resonance sampled at the 1 pm laser step, and Table II summarizes the expected performance. Each stopband contains 4–5 resonances. At λ_B , the FSRs agree with $\lambda_B^2/[2n_g(L + 2L_p)]$ to within 0.3%.

F. Test of the Extraction Equations

MZIs. The spectra of Fig. 4 were sampled at 1 pm. Each transmission minimum was located by a parabolic fit, and Eq. (7) was applied to every adjacent pair. The extracted $n_g(\lambda)$ reproduces the compact model to within 6×10^{-5} for all three MZIs [Fig. 7(a)]. This confirms Eq. (7), including its first-order expansion of β .

FP cavities. The TMM spectra were also sampled at 1 pm, and every resonance inside the stopband was fitted with a Lorentzian. The FSR at λ_B was obtained from the even fit of the resonance spacings [Fig. 7(b)]. Then:

- Eq. (17), with the stopband widths above, gives $n_g = 4.174, 4.173$ and 4.195 for $\Lambda = 307, 312$ and 317 nm. The input values at λ_B are 4.186, 4.188 and 4.191, so the deviation is -0.29% , -0.38% and $+0.11\%$. The error comes from the few resonances inside the stopband and the approximation $L_p \approx 1/2\kappa$. It is comparable to the change in n_g caused by a 2 nm width error.

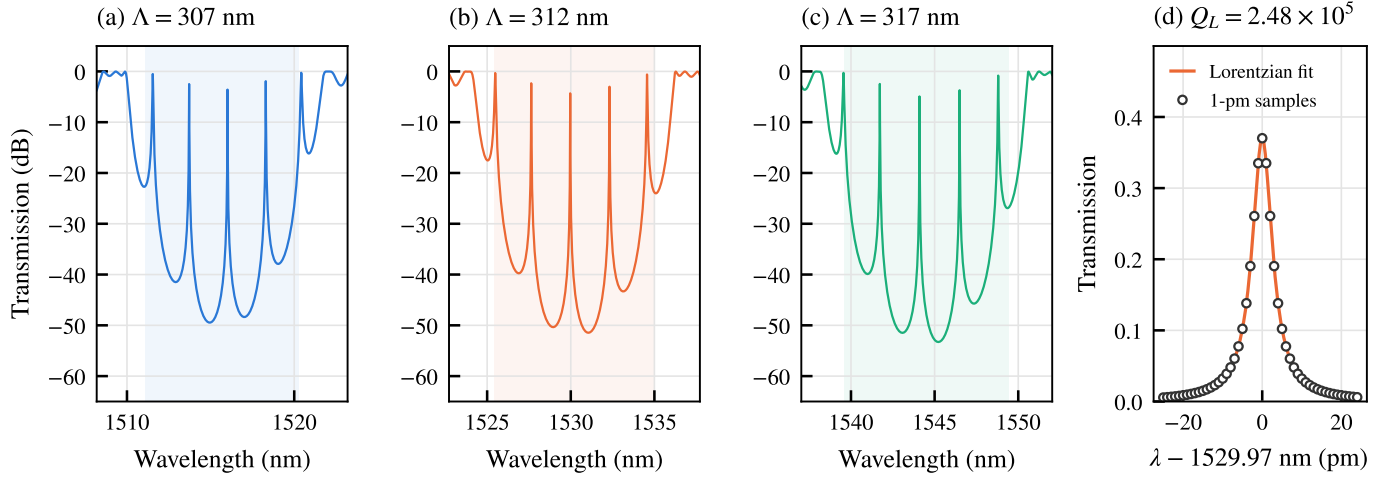


Fig. 6. Simulated transmission of the Bragg FP cavities ($L = 100\mu\text{m}$). (a)–(c) $\Lambda = 307, 312$ and 317 nm; shaded bands mark the FDTD stopbands. (d) The resonance nearest λ_B for $\Lambda = 312$ nm, sampled at 1 pm as in the measurement, with a Lorentzian fit (FWHM 6.2 pm).

TABLE II
DESIGN VARIATIONS AND EXPECTED PERFORMANCE

Device	Type	Λ (nm)	L or ΔL (μm)	λ_B (nm)	$\Delta\lambda_{SB}$ (nm)	R	FSR (nm)	Res. in SB	FWHM (pm)	Q_L	T_{pk} (dB)
MZI1	Y-branch MZI	–	$\Delta L = 50$	–	–	–	$11.46^\dagger$	–	–	–	–
MZI2	Y-branch MZI	–	$\Delta L = 100$	–	–	–	$5.73^\dagger$	–	–	–	–
MZI3	Y-branch MZI	–	$\Delta L = 200$	–	–	–	$2.87^\dagger$	–	–	–	–
FP1	Bragg FP	307	100	1515.7	9.18	0.994	2.311	4	7.0	216 000	–3.6
FP2	Bragg FP	312	100	1530.2	9.54	0.995	2.363	5	6.2	248 000	–4.3
FP3	Bragg FP	317	100	1544.5	9.83	0.996	2.401	4	5.7	271 000	–4.9

All devices: $W = 500\text{ nm}$, $H = 220\text{ nm}$, quasi-TE, $n_g(1550\text{ nm}) = 4.192$. † MZI FSR at 1550 nm from Eq. (7); across 1500 – 1600 nm it varies from 10.76 to 12.19 nm (MZI1), 5.38 to 6.10 nm (MZI2) and 2.69 to 3.05 nm (MZI3). FP cavities: $\Delta W = 40\text{ nm}$, $N = 200$; λ_B , $\Delta\lambda_{SB}$ and κ ($5.26, 5.36, 5.42 \times 10^4\text{ m}^{-1}$) from 3D FDTD; $L_p = 9.5, 9.3$ and $9.2\mu\text{m}$; 1.4 dB/cm propagation loss. R is the mirror reflectivity at λ_B . The FSR is evaluated at λ_B . FWHM, Q_L and T_{pk} refer to the resonance nearest λ_B (Lorentzian fit); T_{pk} is relative to a straight waveguide, excluding the grating couplers.

- With n_g from the MZIs, Eq. (18) gives $L_p = 9.37, 9.14$ and $9.27\mu\text{m}$, hence $\kappa = 5.32, 5.46$ and $5.38 \times 10^4\text{ m}^{-1}$. These agree with the FDTD values ($5.26, 5.36, 5.42 \times 10^4\text{ m}^{-1}$) to within 2%.
- The finesse and Eq. (20), with R from the FDTD κ , give $\alpha = 1.41$ – 1.44 dB/cm (input 1.4 dB/cm).

The measured spectra will be processed in the same way, after the grating-coupler envelope has been removed by a polynomial fit to the MZI fringe maxima or to the out-of-band transmission of the cavities.

IV. FABRICATION

To be completed later to include the layout and details about fabrication.

V. EXPERIMENTAL DATA

To be completed later.

VI. ANALYSIS

To be completed later.

VII. CONCLUSION

To be completed later.

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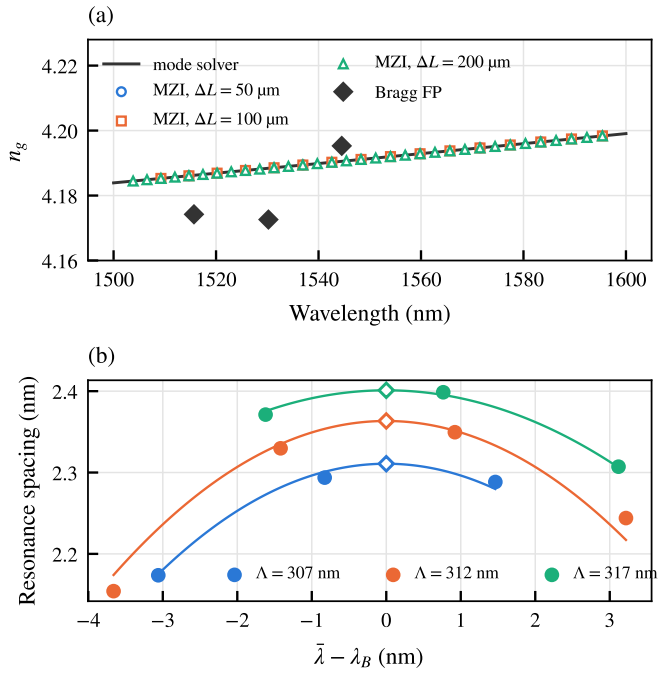


Fig. 7. Test of the group-index extraction on simulated spectra sampled at 1 pm. (a) $n_g(\lambda)$ from the three MZIs [Eq. (7)] and from the three FP cavities [Eq. (17)], compared with the mode solver. (b) Spacing of adjacent FP resonances versus wavelength, with the even fit used to obtain the FSR at λ_B (open diamonds).